UNIT 5 : AI APPLICATIONS

Examples and applications of AI

It can be almost anything. Your smartphone uses AI, as do services like digital assistants, chatbots, social media websites, and much more. Many home electronics also use AI, such as robot vacuum cleaners or security systems. And, of course, there are classic examples of auto-navigation and robotics.

Below, we've outlined applications in greater detail so you can understand how AI impacts everyday life.

Digital assistants

Perhaps the application used by most people would be the digital assistants on our various pieces of technology. If you have a smartphone or laptop, you probably have and use digital assistant software to some degree.

Some of the most popular digital assistants include:

- **Siri (Apple)**
- **Alexa (Amazon)**
- **Cortana (Microsoft)**
- **Google Assistant (Google)**
- **Bixby (Samsung)**

Search engines

Another common application of AI is search engines. Search engine algorithms utilize AI to refine and show better results without the intervention of programmers. You can see this in action on Google if you search a question. You'll see a section called "People also ask" and if you open one of those questions, it will spawn two more related questions below.

An even simpler example is Google's auto-complete answers when you type in the search bar. An AI algorithm gathers data on what people search most often and uses that to populate predictions you can use to navigate.

Popular search engines include:

- **Google**
- **Yahoo**
- **Bing**
- **DuckDuckGo**

Social media

Social media platforms are another common way people interact with AI. All major social media platforms run off AI-powered algorithms which are designed to serve specific purposes. Most use algorithms to determine what their users like and serve more of that content, to keep the user engaged. Many also run AI algorithms to gather and store user data to use for advertising purposes.

You can train your social media algorithms to show the content you like by creating filters, or searching carefully for what you like, and purposefully interacting (liking, commenting, sharing, etc.) with things you enjoy.

Popular social media platforms include:

- **Facebook (Meta)**
- **Instagram (Meta)**
- **YouTube**
- **TikTok**

Online shopping

This is probably one of the least obvious ways people interact with AI in their daily lives. Many online shopping and ecommerce platforms use AI to streamline their customer experience in a variety of ways.

As a customer, you may experience AI through:

- **Personalized product recommendations based on previous shopping activity or customer profile.**
- **Pricing optimization based on supply, demand, or previous shopping activity.**
- **Chatbots to provide instant responses to customer service or technical issues.**

- Shipping and delay estimates.

As a business owner, you may consider implementing AI in the following additional ways:

- Sales and demand forecasting to help you manage your inventory in the face of increased or decreased demand.
- Creating customer profiles and segmentation to boost sales.
- Smart analytics to show in real-time how your business is performing.

Robots

The word "robot" probably makes many people think of sci-fi movies like Star Wars or shows like Star Trek with their humanoid, intelligent robots. Though those may seem futuristic or even far-fetched, in reality, many robots already exist in our world. You may even own some, or something produced by one.

Robots are used in a myriad of fields to streamline production or keep workers safe. They handle repetitive tasks or anything deemed too dangerous for a human worker. Some examples of industrial robots include:

- Aerospace: You may be familiar with the Mars rovers NASA has landed over the years. These are programmed to explore, gather samples and send transmissions back to Earth to provide data from Mars that an astronaut would be unable to obtain. Most recently, NASA sent the rover Perseverance to Mars to gather samples and search for signs of ancient life.
- Manufacturing: The use of robots in assembly lines dates back to 1961, when General Motors introduced a robot to assist with welding and transporting die casings (jobs deemed too dangerous for humans). It continues to this day, streamlining production and providing safer working conditions for humans.
- Hospitality: Particularly in recent years, the hospitality industry has adopted robots to help complete simple tasks and fill in for worker shortages. These can do things like check-in guests at hotels, mix drinks at cafes, deliver meals to tables in restaurants, and more.

Language Model in Artificial Intelligence

Artificial intelligence language models are used to understand, generate and manipulate human-like responses. They are sophisticatedly designed to understand patterns, syntax, semantics, and context of language by training on massive textual datasets. Language Models in NLP (Natural language processing) allow computers to facilitate various language-related activities to interpret, comprehend, and produce text that resembles human speech .

Large Language Model (LLM Model)

A large language model in AI is a type of sophisticated language model that is developed to comprehend and produce human language using vast volumes of text data. They are distinguished by their immense size, capability to handle complicated language structures, and capacity to create coherent and contextually relevant text. In contrast to standard language models in NLP, which may be constrained in their breadth and correctness.

LLM models can capture a variety of linguistic subtleties and patterns since they include billions of parameters. It makes them powerful tools in applications that require sophisticated language understanding and generation.

The concept of language modeling in NLP is all about teaching machines to communicate like humans do, enabling businesses to streamline customer interactions through chatbots, voice assistants, and other channels. With advanced language models, companies can automate routine inquiries, provide personalized responses, and enhance operational efficiency. Language models are revolutionizing customer engagement, promising a future where communication is seamless and customer experiences are exceptional.

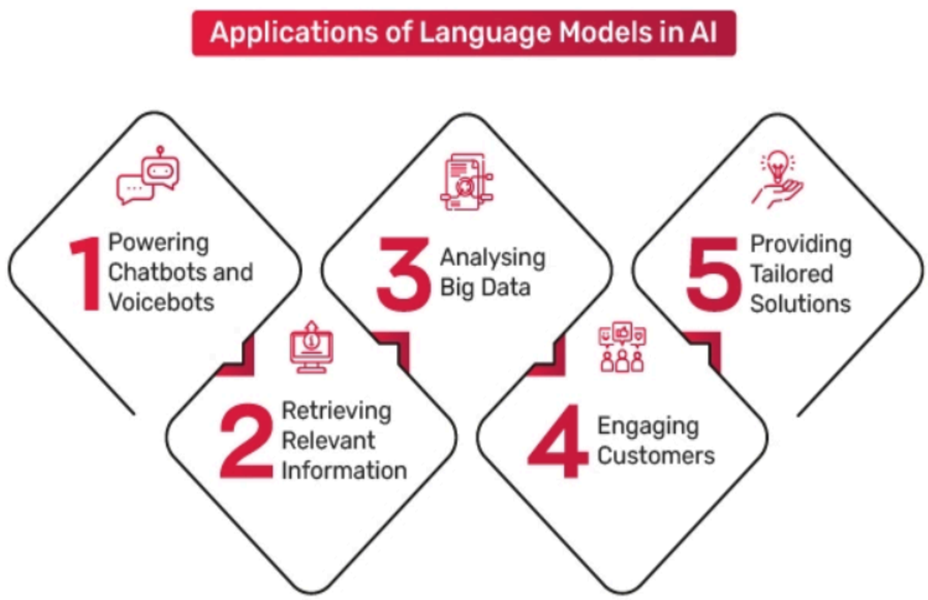
Language Modeling Types

By leveraging advanced machine learning techniques, language models enable businesses to automate and streamline their customer interactions, ultimately driving efficiency and improving customer satisfaction. In this section, we will explore the different types of language models that are revolutionizing the way businesses engage with their customers.

- **Rule-based Language Models:** Rule-based language models follow a predefined set of linguistic rules to generate responses. While they are simple and straightforward, they lack the ability to understand context and may produce generic and rigid responses.
- **Statistical Language Models:** Statistical language models utilize statistical techniques to generate responses based on the frequency of word sequences in training data. While they have improved contextual understanding compared to rule-based models, they still struggle with the ambiguity and complexity of language.
- **Neural Language Models:** Neural language models, powered by deep learning algorithms, have ushered in a new era of customer interactions. These models are trained on vast amounts of data, allowing them to capture complex language patterns and generate highly accurate and contextually relevant responses. Neural language models leverage techniques such as recurrent neural networks (RNNs) and transformers to achieve state-of-the-art performance.
- **Generative Pre-trained Transformers:** Generative pre-trained transformers, or GPT models, have gained significant attention in recent years. These models are pre-trained on large amounts of diverse data, enabling them to generate human-like responses with unprecedented accuracy. GPT models excel in understanding context and generating coherent and contextually appropriate responses.

By leveraging these advanced language modeling techniques, businesses can automate customer interactions across various channels, such as chatbots, voice assistants, and email automation. This not only saves time and resources but also ensures consistent and personalized customer experiences.

Applications of Language Model in Artificial Intelligence



Language models are a stepping stone towards general AI, enabling machines to understand, process, and generate human-like language, thereby improving human-machine interaction. LMs convert qualitative data into quantitative data for devices and computers to comprehend. Here are some applications of the language model in an AI application:

1. **Chatbots and Voicebots:** Language models power chatbots and voicebots that can engage in natural language conversations with users. It enables computers to understand and generate human language & assisting them with queries, providing information, language translation, sentiment analysis, and more.
2. **Information Retrieval:** These models refine search engines, helping users find relevant information faster and more accurately by understanding their queries better.

3. **Data Analysis:** Language models assist in extracting insights from unstructured text data, enabling businesses to gain valuable information from customer feedback, social media, and reviews.
4. **Customer Engagement:** In customer service, language models drive automated responses that offer quick solutions, enhancing customer satisfaction and saving time.
5. **Personalization:** Language models enable companies to tailor content, recommendations, and offers to individual preferences, increasing user engagement and conversion rates.

Language models in artificial intelligence have various applications across various fields due to its transformative impact on understanding, communication, and automation in the digital world. Here are some applications of language models in different industries:

1. **Education:** They aid in educational platforms by generating interactive content, quizzes, and explanations, making learning more engaging and effective.
2. **Medical Diagnostics:** They assist in analysing medical literature, assisting doctors in diagnosing and researching diseases.
3. **Language Preservation:** They can help preserve and revitalise endangered languages by analysing existing texts and generating content.
4. **Legal Documentation:** In the legal field, the models help draft contracts, analyse case law, and generate legal documents.
5. **Market Research:** Businesses can analyse online conversations and discussions to gauge public sentiment and trends, helping refine marketing strategies.

Examples of Language Models

Language models have emerged as a powerful tool, revolutionizing how we interact with technology. Here are several notable examples:

- **GPT-3 (Generative Pre-trained Transformer 3):** Developed by OpenAI, GPT-3 stands out as a cutting-edge language model with a staggering 175 billion parameters. Its exceptional capacity enables it to generate human-like text, perform language translation, answer questions, compose music, and even create code. The versatility of GPT-3 makes it invaluable for a wide range of applications.
- **BERT (Bidirectional Encoder Representations from Transformers):** Powered by transformer architecture, BERT excels at understanding context within textual data. It has been successfully employed in sentiment analysis, question answering, and language translation. BERT's contextual understanding and extensive training on vast amounts of text contribute to its remarkable effectiveness in natural language processing tasks.
- **ELMO (Embeddings from Language Models):** ELMO introduced contextual embeddings and significantly improved word representations. By considering surrounding words and context, ELMO generates nuanced and accurate word embeddings that enhance performance in language understanding tasks, such as sentiment analysis, machine translation, and document classification.

Future of Language Modeling

The future of language modeling holds great promise for multi-modal interactions. With the proliferation of voice assistants, smart devices, and chatbots, language models are evolving to support seamless interactions across different channels. This enables businesses to provide consistent and cohesive experiences, regardless of the customer’s preferred mode of communication, whether it be voice, text, or even visual interfaces.

One of the key advancements shaping the future of language modeling is the rise of contextually aware models. With the ability to understand nuanced context, these models go beyond generic responses to provide personalized and tailored interactions. By analyzing vast amounts of data and leveraging sophisticated algorithms, these models enable businesses to create conversational agents that can truly understand customer needs and deliver meaningful solutions.

INFORMATION RETRIEVAL

Information retrieval (IR) is finding material (usually documents) of an unstructured nature (usually text) that satisfies an information need from within large collections (usually stored on computers).Generically, —collections‖, Less-frequently used, —corpora‖ are searched and —documents‖ namely web pages, PDFs, PowerPoint slides, paragraphs, etc. are retrieved. Information Retrieval system consists of a software program that facilitates a user in finding the information the user needs. The Information Retrieval System was coined by Calvin Mooers in 1952. These information retrieval systems were, truly speaking, document retrieval system, since they were designed to retrieve information. Information retrieval deals with storage, organization and access to text, as well as multimedia information resources. Information Retrieval is a process of searching some collection of documents, using the term document in its widest sense, in order to identify those documents which deal with a particular subject. Any system that is designed to facilitate this literature searching may legitimately be called an information retrieval system. Information retrieval systems originally meant text retrieval systems, since they were dealing with textual documents, modern information retrieval systems deal with multimedia information comprising text, audio, images and video. While many features of conventional text retrieval system are equally applicable to multimedia information retrieval, the specific nature of audio, image and video information have called for the development of many new tools and techniques for information retrieval. Modern information retrieval deals with storage, organization and access to text, as well as multimedia information resources. The concept of information retrieval presupposes that there are some documents or records containing information that have been organized in an order suitable for easy retrieval. The documents or records we are concerned with contain bibliographic information which is quite different from other kinds of information or data. We may take a simple example. If we have a database of information pertaining to an office, or a supermarket, all we have are the different kinds of records and related facts, like names of employees, of employees, their positions, salary, and so on, or in the case of a supermarket, names of different items, prices, quantity, and so on. The main objective of a bibliographic information retrieval system, however, is to retrieve the information either the actual information or the documents containing the information that fully or partially match the user_s query. The database may contain abstracts or full texts of document, like newspaper articles, handbooks, dictionaries, encyclopedias, legal documents, statistics, etc., as well as audio, images, and video information.

An information retrieval system thus has three major components- the document subsystem, the users subsystem, and the searching/retrieval subsystem. These divisions are quite
broad and each one is designed to serve one or more functions, such as:

- Analysis of documents and organization of information (creation of a document database)
- Analysis of user_s queries, preparation of a strategy to search the database
- Actual searching or matching of users queries with the database, and finally
- Retrieval of items that fully or partially match the search statement.

An IR is a 3 step Process:

- Asking a question (how to use the language to get what we want?)
- Building an answer from known data. (How to refer to a given text?)
- Assessing the answer. (Does it contain the information we are seeking.)

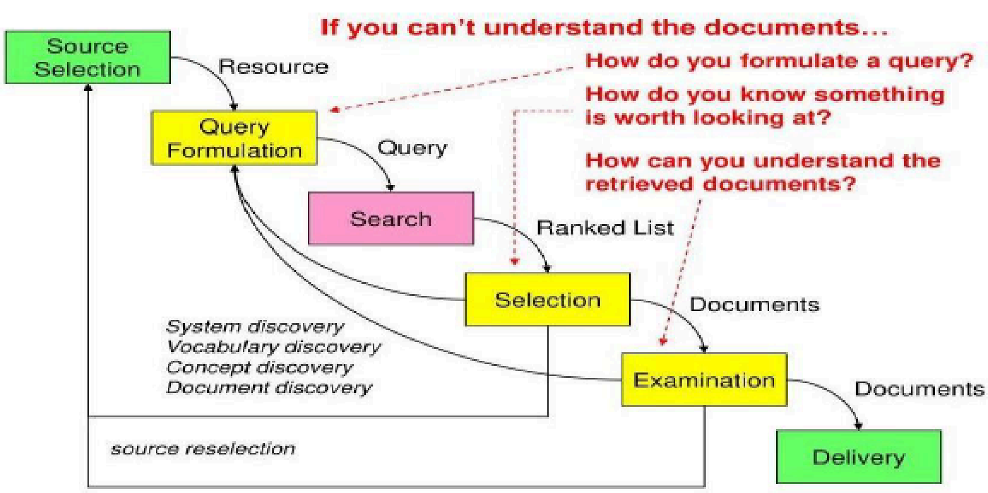
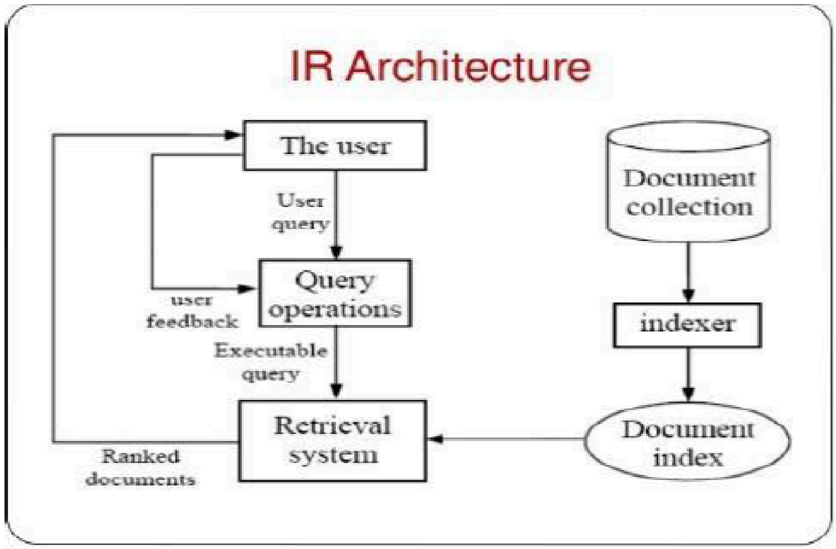


Fig: The Information Retrieval Cycle



5.1.1 IR System Components

- Text Operations forms index words (tokens).
  - ✓ Stop word removal
  - ✓ Stemming
- Indexing constructs an inverted index of word to document pointers.
- Searching retrieves documents that contain a given query token from the inverted index.
- Ranking scores all retrieved documents according to a relevance metric.
- User Interface manages interaction with the user:
  - ✓ Query input and document output.
  - ✓ Relevance feedback.
  - ✓ Visualization of results.
- Query Operations transform the query to improve retrieval:
  - ✓ Query expansion using a thesaurus.
  - ✓ Query transformation using relevance feedback.

5.1.2 Purpose of Information Retrieval System

An information retrieval system is designed to retrieve the documents or information required by the user community. It should make the right information available to the right user. Thus, an information retrieval system aims at collecting and organizing information in one or more subject areas in order to provide it to the user as soon asked for. Belkin presents the following situation which clearly reflects the purpose of information retrieval systems:

- A writer presents as set of ideas in a document using a set of concepts
- Somewhere there will be some users who require the ideas but may not be able to identify those. In other words, there will be some persons who lack the ideas put forward by the author in his/her work.
- Information retrieval system serve to match the writers ideas expressed in the document with the user requirements or demand for those.
- Thus, an information retrieval system serves as a bridge between the world of creators or generators of information and the users of that information.

Some terminology

- An IR system looks for data matching using some criteria defined by the users in their queries.
- The language used to ask a question is called the query language.
- These queries use keywords (atomic items characterizing some data).
- The basic unit of data is a document (can be a file, an article, a paragraph, etc.).
- A document corresponds to free text (may be unstructured).
- All the documents are gathered into a collection (or corpus).

Example:

1 million documents, each counting about 1000 words

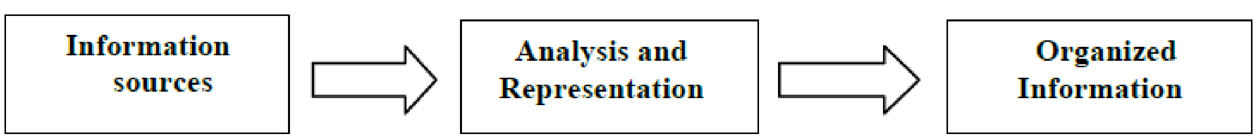
if each word is encoded using 6 bytes:

$109 \times 1000 \times 6/1024 \simeq 6GB$

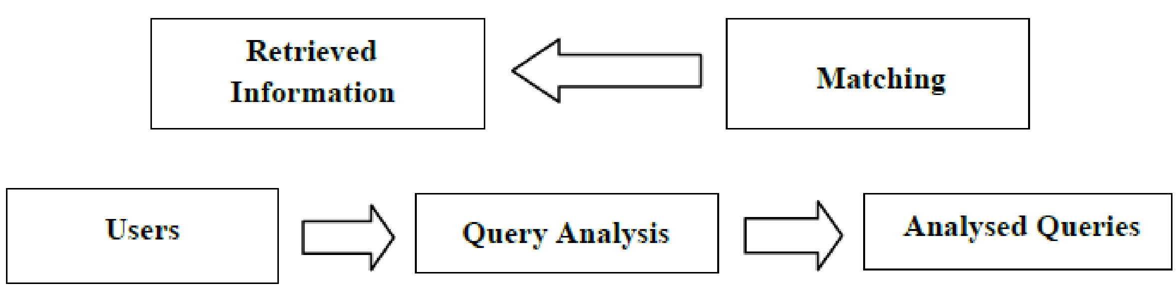
5.1.3 Components of Information Retrieval

In an information retrieval system there are the documents or sources of information on one side and on the other there are the user_s queries. These two sides are linked through a series of tasks. Lancaster mentions that an information retrieval system comprises six major subsystems: The document subsystem

- ✓ The indexing subsystem
- ✓ The vocabulary subsystem
- ✓ The searching subsystem
- ✓ The service-system interface, and
- ✓ The matching subsystem



The broad outline of an information retrieval system



Three major components of IRS

1) Document subsystem

    a) Acquisition

    b) Representation

    c) File organization

2) User sub system

    a) Problem

    b) Representation

    c) Query

3) Searching /Retrieval subsystem

    a) Matching

    b) Retrieved objects

5.1.4 Kinds of Information Retrieval Systems

Two broad categories of information retrieval system can be identified: in- house and online.

In- house information retrieval systems are set up by a particular library or information center to serve mainly the users within the organization. One particular type of in-house database is the library catalogue. Online public access catalogues (OPACs) provide facilities for library users to carry out online catalogue searches, and then to check the availability of the item required. Online IR is nothing but retrieving data from web sites, web pages and servers that may include data bases, images, text, tables, and other types.

5.3.4 Functions of information retrieval system

An information retrieval system deals with various sources of information on the one hand and user_s requirements on the other. It must:

- ✓ Analyze the contents of the sources of information as well as the user_s queries, and then
- ✓ Match these to retrieve those items that are relevant

The major functions of an information retrieval system can be listed as follows:

- ✓ To identify the information (sources) relevant to the areas of interest of the target users community
- ✓ To analyze the contents of the sources (documents)
- ✓ To represent the contents of the analyzed sources in a way that will be suitable for matching user_s queries
- ✓ To analyze user_s queries and to represent them in a form that will be suitable for matching with the database
- ✓ To match the search statement with the stored database
- ✓ To retrieve the information that is relevant, and
- ✓ To make necessary adjustments in the system based on feedback form the users.

5.3.5 Features of an information retrieval system

- ✓ An effective information retrieval system must have provisions for:
- ✓ Prompt dissemination of information
- ✓ Filtering of information
- ✓ The right amount of information at the right time
- ✓ Active switching of information
- ✓ Receiving information in an economical way
- ✓ Browsing
- ✓ Getting information in an economical way
- ✓ Current literature
- ✓ Access to other information systems
- ✓ Interpersonal communications, and
- ✓ Personalized help.

5.3.6 Indexing usually consists of the several phases

- ✓ After word segmentation, stop words are removed.
- ✓ These common words like articles or prepositions contain little meaning by themselves and are ignored in the document representation.
- ✓ Second, word forms are transformed into their basic form, the stem.
- ✓ During the stemming phase, e.g. houses would be transformed into house.
- ✓ For the document representation, different word forms are usually not necessary.
- ✓ The importance of a word for a document can be different.
- ✓ Some words better describe the content of a document than others.
- ✓ This weight is determined by the frequency of a stem within the text of a document.

In multimedia retrieval, the context is essential for the selection of a form of query and document representation. Different media representations may be matched against each other or transformations may become necessary (e.g. to match terms against pictures or spoken language utterances against documents in written text).

As information retrieval needs to deal with vague knowledge, exact processing methods are not appropriate.

- Vague retrieval models like the probabilistic model are more suitable.
- Within these models, terms are provided with weights corresponding to their importance for a document.
- These weights mirror different levels of relevance.

The result of current information retrieval systems are usually sorted lists of documents where the top results are more likely to be relevant according to the system.

- In some approaches, the user can judge the documents returned to him and tell the systems which ones are relevant for user.
- The system then resorts the result set.
- Documents which contain many of the words present in the relevant documents are ranked higher.
- This relevance feedback process is known to greatly improve the performance.
- Relevance feedback is also an interesting application for machine learning.
- Based on a human decisions, the optimization step can be modeled with several approaches, e.g. with rough sets.

5.2 INFORMATION EXTRACTION

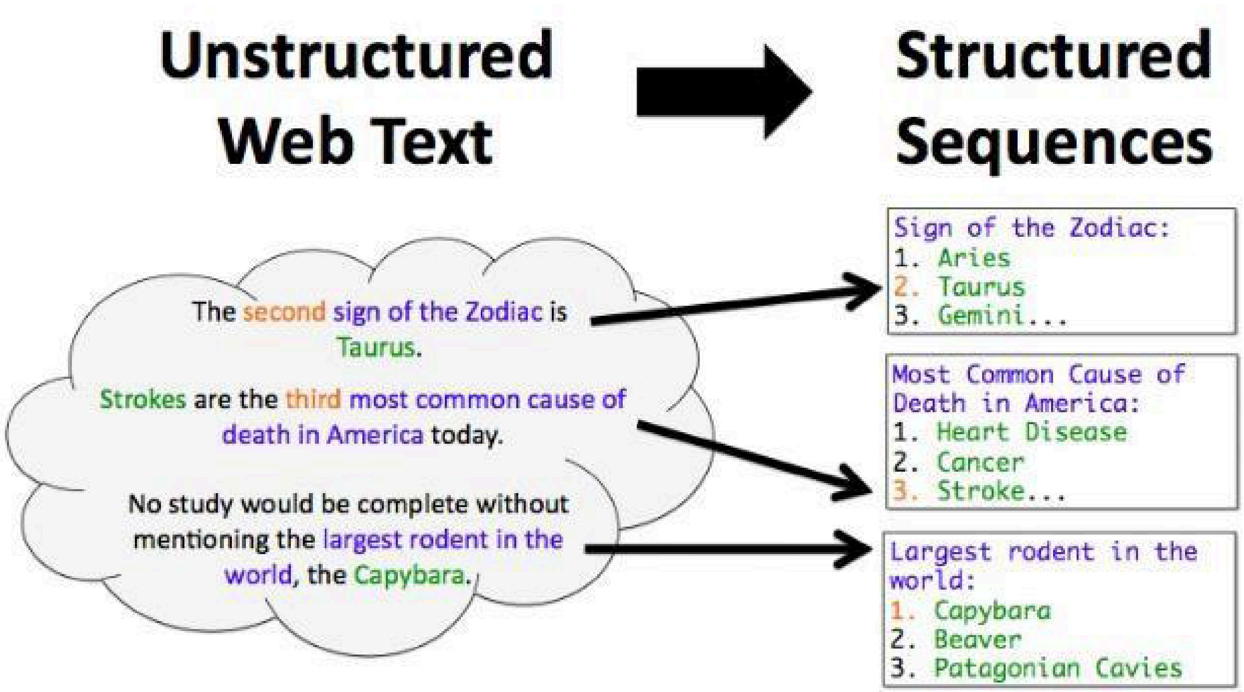
Information extraction (IE) is the automated retrieval of specific information related to a selected topic from a body or bodies of text. Information extraction is the process of extracting specific (pre-specified) information from textual sources. One of the most trivial examples is when your email extracts only the data from the message for you to add in your Calendar.

Other free-flowing textual sources from which information extraction can distill structured information are legal acts, medical records, social media interactions and streams, online news, government documents, corporate reports and more.

Information extraction tools make it possible to pull information from text documents, databases, websites or multiple sources. IE may extract info from unstructured, semi-structured or structured, machine-readable text. Usually, however, IE is used in natural language processing (NLP) to extract structured from unstructured text.

Information extraction depends on,

- Named entity recognition (NER), a sub-tool used to find targeted information to extract.
- NER recognizes entities first as one of several categories such as location (LOC), persons (PER) or organizations (ORG).
- Once the information category is recognized, an information extraction utility extracts the named entity_s related information and constructs a machine-readable document from it, which algorithms can further process to extract meaning.
- IE finds meaning by way of other subtasks including co-reference resolution, relationship extraction, language and vocabulary analysis and sometimes audio extraction.
- Current efforts in multimedia document processing in IE include automatic annotation and content recognition and extraction from images and video could be seen as IE as well.
- Because of the complexity of language, high-quality IE is a challenging task for artificial intelligence (AI) systems.



Typically, for structured information to be extracted from unstructured texts, the following main subtasks are involved:

Pre-processing of text – where text is prepared for processing with the help of computational linguistics tools such as tokenization, sentence splitting, morphological analysis, etc.

Finding and classifying concepts – this is where mentions of people, things, locations, events and other pre-specified types of concepts are detected and classified.

Connecting the concepts – task of identifying relationships between extracted concepts.

Unifying – this subtask is about presenting the extracted data into a standard form.

Getting rid of the noise – this subtask involves eliminating duplicate data.

Enriching your knowledge base – this is where the extracted knowledge is ingested in your database for further use.

5.2.1 Information Extraction Architecture

The below figure shows the architecture for a simple information extraction system. At first, the raw text of the document is split into sentences using a sentence segmenter, and each sentence is further subdivided into words using a tokenizer. Next, each sentence is tagged with part-of-speech tags, which will prove very helpful in the next step, **named entity detection**. In this step, we search for mentions of potentially interesting entities in each sentence. Finally, we use **relation detection** to search for likely relations between different entities in the text.

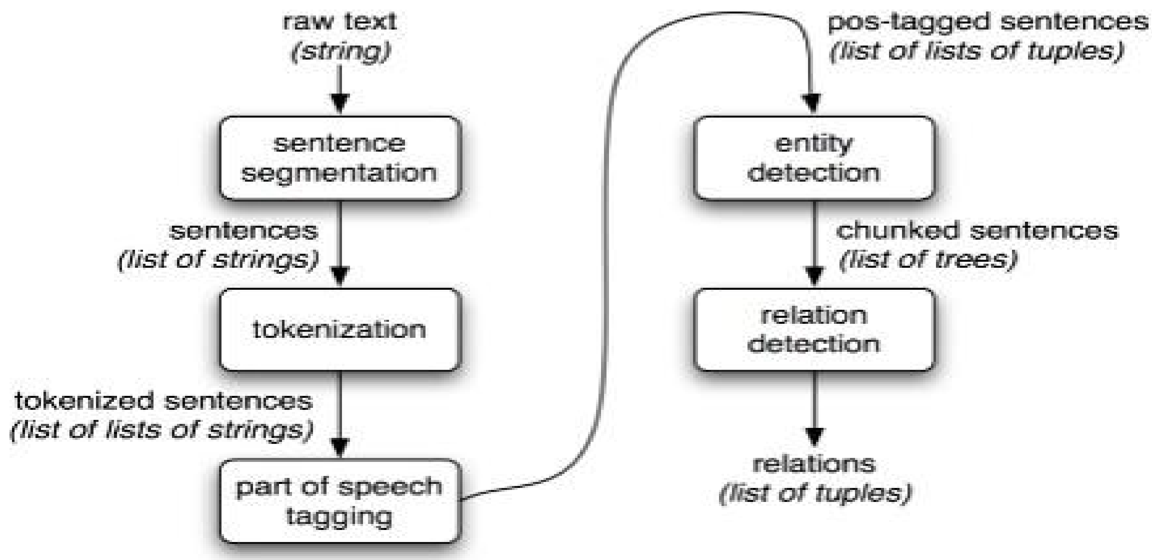


Figure : Simple Pipeline Architecture for an Information Extraction System.

This system takes the raw text of a document as its input, and generates a list of (entity, relation, entity) tuples as its output.

5.2.2 Applications of IE

- Enterprise
- News tracking
- Customer care
- Data cleaning
- Personal information management
- Scientific applications
- Web oriented applications
- Citation databases
- Opinion databases
- Community websites
- Comparison shopping
- Ad placement on webpages
- Structured web searches

Artificial Intelligence is the ability of machines to perform tasks like thinking, reasoning, and learning similar to humans. It is a broad field in science that includes various techniques like Machine learning, Natural Language Processing, Robotics, and more.

What is Machine Learning?

Machine learning (ML) is a subset of artificial intelligence that enables machines to learn from data without being explicitly programmed. It uses algorithms to analyze large amounts of data, learn from the insights, and gain patterns and make informed decisions.

In simple words, the machine "learns" from the data and uses this knowledge to make predictions and decisions.
Machine learning algorithms enhance their performance over time as they undergo continuous training and exposed to additional data. Machine learning models are the output or what the program learns by executing an algorithm on training data. The greater the amount of data used, the better the model will get.

There are three main types of machine learning −

- Supervised Learning: In this type of learning, the machine is given labeled data to train algorithms especially to classify data or predict outcomes. Some of the algorithms include Linear regression, Logistic regression, Random Forest, and other.
- Unsupervised Learning: In this type of learning, the machine is given unlabeled datasets to algorithms to find hidden patterns or data groupings. There are three types of unsupervised learning tasks which include clustering, association rule, and dimensionality reduction.
- Reinforcement Learning: In this type of learning, an agent is trained to interpret the environment and learns from the feedback which is can be a reward (positive feedback) or penalty (negative feedback).

Artificial Intelligence and Machine Learning are two term that are often used interchangeably. However, they are not the same thing but are closely connected.

Relationship Between AI and ML

Understanding the relationship between AI and ML is important for developing intelligent systems. The simplest way to understand this is −

- AI is the **broader concept** of enabling a system to think, act, and learn like humans.
- ML is an **application of AI** that allows machines to learn from data and gain knowledgeable insights.

Artificial Intelligence is the branch of computer science that covers a variety of approaches and algorithms, and machine learning being one of it.

Machine Learning Vs Artificial Intelligence

We are often confused between machine learning and artificial intelligence. The table below consists of the difference between both the terms −

Aspect	**Artificial Intelligence (AI)**	**Machine Learning (ML)**
Definition	AI refers to the enabling systems to think, act, and learn similar to humans.	ML is a subset of AI that focuses on algorithms that learn from data and gain insights.
Scope	AI is a broad field that consists of various other technologies like **Natural Language Processing** and **Robotics**.	
Goals	To create systems that can perform tasks replicating humans.	To develop models that can improve their performance over time as they are exposed to more data.
Techniques Used	Includes reasoning, learning, planning, and understanding.	Primarily uses statistical methods, neural networks, and decision trees.

Comple xity	Often involves multiple systems and layers of abstraction.	Generally focuses on specific tasks and can be less complex.

Benefits of using AI and ML Together

AI and ML together offer some key benefits to enhance organizations and businesses which include −

- **Wider Data Ranges:** Analyzing and understanding a wide range of structured and unstructured data sources.
- **Better Decision-Making:** Improving data integrity, accelerating data processing, and reducing human error for more better and relevant decision-making.
- **Efficiency:** Increasing operational efficiency and reducing costs.
- **Analytic Integration:** Integrating predictive analytics and insights into business help them grow.

Natural Language Processing

Natural Language Processing (NLP)refers to AI method of communicating with an intelligent systems using a natural language such as English. Processing of natural language is required when you want an intelligent system like robot to perform as per your instructions, when you want to hear decision from a dialogue based clinical expert system, etc.
The field of NLP involves making computers to perform useful tasks with the natural languages humans use. The input and output of an NLP system can be **speech or a written text**.

Key Terms in NLP

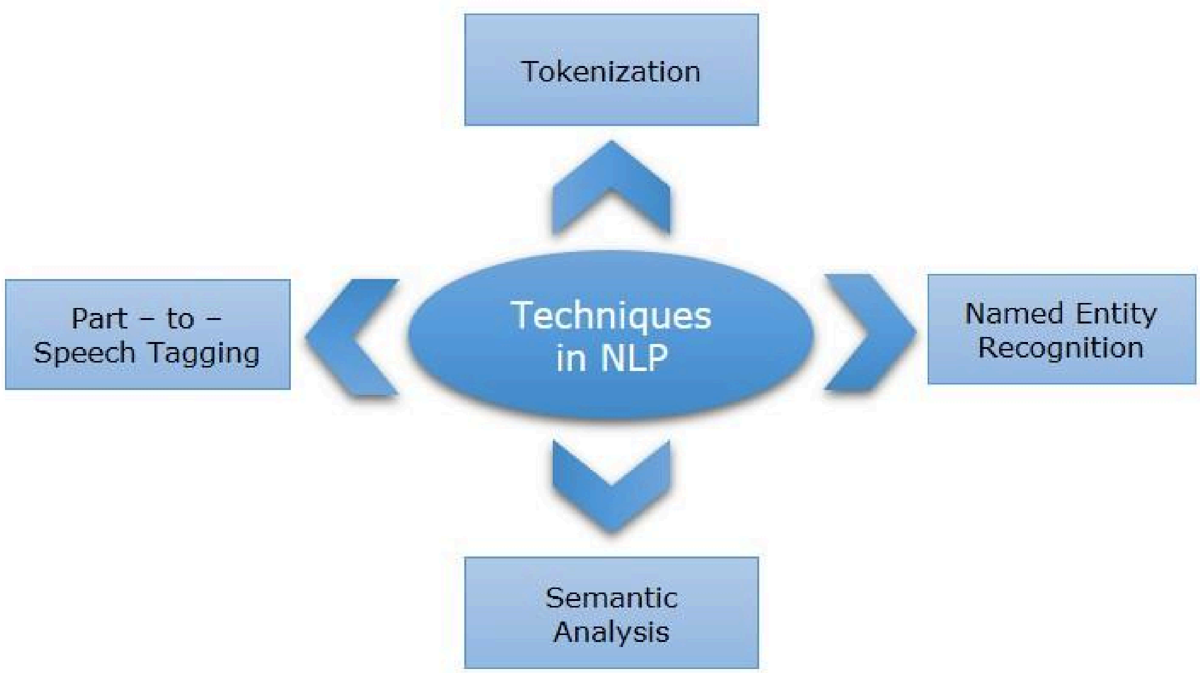
Some of the key terms and concepts in natural language processing are listed below −

- **Phonology** − It is study of organizing sound systematically.
- **Morphology** − It is a study of construction of words from primitive meaningful units.
- **Morpheme** − It is primitive unit of meaning in a language.

Syntax − It refers to arranging words to make a sentence. It also involves determining the structural role of words in the sentence and in phrases.
Semantics − It is concerned with the meaning of words and how to combine words into meaningful phrases and sentences.
Pragmatics − It deals with using and understanding sentences in different situations and how the interpretation of the sentence is affected.
Discourse − It deals with how the immediately preceding sentence can affect the interpretation of the next sentence.
World Knowledge − It includes the general knowledge about the world.

Techniques in NLP

Techniques in NLP are methods and algorithms used to process, analyze, and understand human language and data. Some of the common NLP techniques are −



Tokenization − It is a technique in NLP that involves splitting a sentence or phrase into smaller units known as tokens.

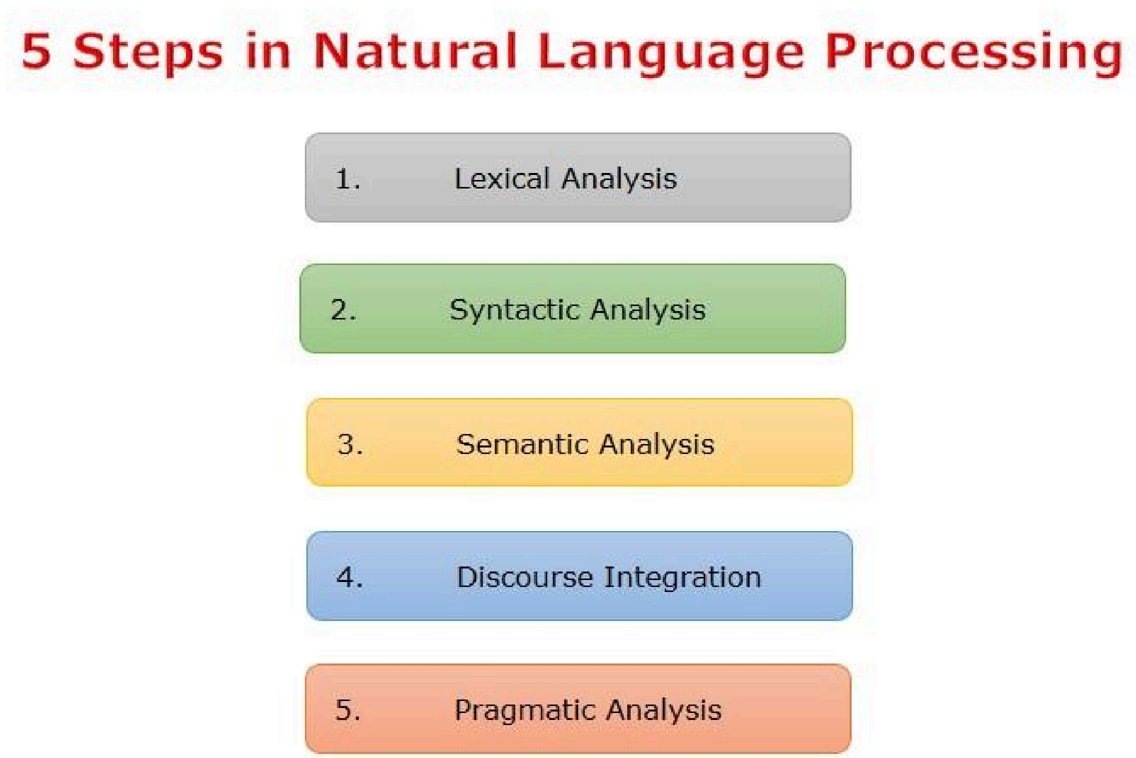
Part-to-Speech Tagging − This technique in NLP involves the process of identifying and labeling works in a sentence based in their part of speech (noun, verb, adjective).

Named Entity Recognition (NER) − This technique in NLP is used to identify named entities in the text, such as people, organizations, locations, dates, and more.

Semantic Analysis − This technique in NLP determines the sentiment expressed in a piece of text.

Steps in NLP

For the better understanding, analysis of written and spoken language effectively the following 5 NLP steps are followed



Lexical Analysis − It involves identifying and analyzing the structure of words. Lexicon of a language means the collection of words and phrases in a language. Lexical analysis is dividing the whole chunk of text into paragraphs, sentences, and words.

Syntactic Analysis (Parsing) − It involves analysis of words in the sentence for grammar and arranging words in a manner that shows the relationship among the words. The sentence such as â œThe school goes to boyâ is rejected by English syntactic analyzer.

Semantic Analysis − It draws the exact meaning or the dictionary meaning from the text. The text is checked for meaningfulness. It is done by mapping syntactic structures and objects in the task domain. The semantic analyzer disregards sentence such as â œhot ice-creamâ.

Discourse Integration − The meaning of any sentence depends upon the meaning of the sentence just before it. In addition, it also brings about the meaning of immediately succeeding sentence.

Pragmatic Analysis − During this, what was said is re-interpreted on what it actually meant. It involves deriving those aspects of language which require real world knowledge.

Components of NLP

There are two components of NLP as given −

Natural Language Understanding (NLU)

NLU helps machine to understand and analyse human language by extracting metadata from content which includes concepts, entities, keywords, emotion, relations, and semantic roles. Understanding involves the following tasks −

Mapping the given input in natural language into useful representations.

Analyzing different aspects of the language.

Natural Language Generation (NLG)

It is the process of producing meaningful phrases and sentences in the form of natural language from some internal representation.

It involves −

Text planning − It includes retrieving the relevant content from knowledge base.

Sentence planning − It includes choosing required words, forming meaningful phrases, setting tone of the sentence.

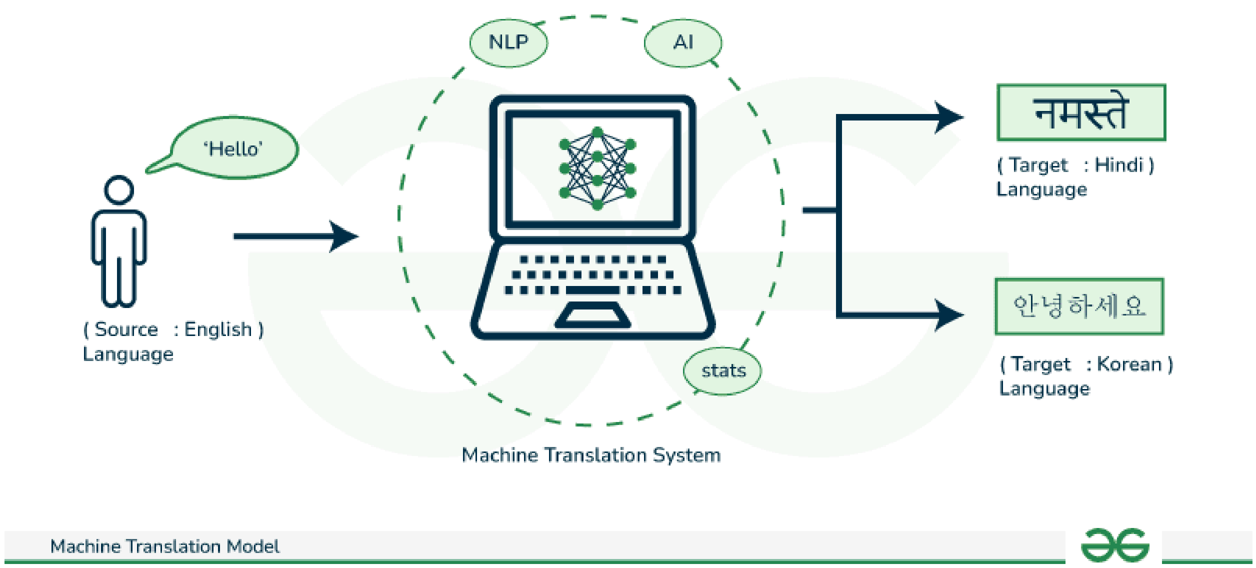
Text Realization − It is mapping sentence plan into sentence structure.

Machine Translation in AI

Machine translation of languages refers to the use of **artificial intelligence (AI)** and machine learning algorithms to automatically translate text or speech from one language to another. This technology has been developed over the years and has become increasingly sophisticated, with the ability to produce accurate translations across a wide range of languages. The article explores more about machine translation, why we need it and its applications.

What is Machine Translation?

Machine translation is a sub-field of computational linguistics that focuses on developing systems capable of automatically translating text or speech from one language to another. In Natural Language Processing (NLP), the goal of machine translation is to produce translations that are not only grammatically correct but also convey the meaning of the original content accurately.



History of Machine Translation

The automatic translation of text from one natural language (the source) to another is known as machine translation (the target). It was one of the first applications for computers that were imagined (Weaver, 1949).

There have been three primary uses of machine translation in the past:

1. Rough translation, such as that given by free internet services, conveys the "gist" of a foreign statement or document but is riddled with inaccuracies. Companies utilize pre-edited translation to publish documentation and sales materials in several languages.
2. The original source content is written in a limited language that makes machine translation easier, and the outputs are often edited by a person to rectify any flaws.
3. Restricted-source translation is totally automated, but only for highly stereotyped language like a weather report.

What are the key approaches in Machine Translation?
In machine translation, the original text is decoded and then encoded into the target language through two step process that involves various approaches employed by language translation technology to facilitate the translation mechanism.

1. Rule-Based Machine Translation

Rule-based machine translation relies on these resources to ensure precise translation of specific content. The process involves the software parsing input text, generating a transitional representation, and then converting it into the target language with reference to grammar rules and dictionaries.

2. Statistical Machine Translation

Rather than depending on linguistic rules, statistical machine translation utilizes machine learning for text translation. Machine learning algorithms examine extensive human translations, identifying statistical patterns.

When tasked with translating a new source text, the software intelligently guesses based on the statistical likelihood of specific words or phrases being associated with others in the target language.

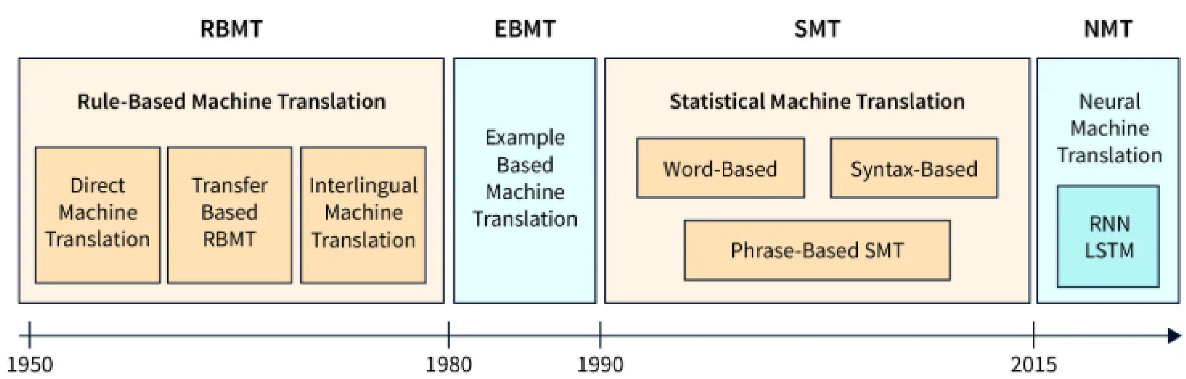
3. Neural Machine Translation (NMT)

A neural network, inspired by the human brain, is a network of interconnected nodes functioning as an information system. Input data passes through these nodes to produce an output. Neural machine translation software utilizes neural networks to process vast datasets, with each node contributing a specific change from source text to target text until the final result is obtained at the output node.

4. Hybrid Machine Translation

Hybrid machine translation tools integrate multiple machine translation models within a single software application, leveraging a combination of approaches to enhance the overall effectiveness of a singular translation model. This process typically involves the incorporation of rule-based and statistical machine translation subsystems, with the ultimate translation output being a synthesis of the results generated by each subsystem.

Evolution of Machine Translation



Why we need Machine Translation in NLP?

Machine translation in Natural Language Processing (NLP) has several benefits, including:

1. **Improved communication:** Machine translation makes it easier for people who speak different languages to communicate with each other, breaking down language barriers and facilitating international cooperation.
2. **Cost savings:** Machine translation is typically faster and less expensive than human translation, making it a cost-effective solution for businesses and organizations that need to translate large amounts of text.
3. **Increased accessibility:** Machine translation can make digital content more accessible to users who speak different languages, improving the user experience and expanding the reach of digital products and services.
4. **Improved efficiency:** Machine translation can streamline the translation process, allowing businesses and organizations to quickly translate large amounts of text and improving overall efficiency.
5. **Language learning:** Machine translation can be a valuable tool for language learners, helping them to understand the meaning of unfamiliar words and phrases and improving their language skills.

What is the application of Machine Translation?
Machine translation has many applications, including:

1. **Cross-border communication:** Machine translation allows people from different countries to communicate with each other more easily, breaking down language barriers and facilitating international cooperation.
2. **Localization:** Machine translation can be used to quickly and efficiently translate websites, software, and other digital content into different languages, making them more accessible to users around the world.
3. **Business:** Machine translation can be used by businesses to translate documents, contracts, and other important materials, enabling them to work with partners and customers from around the world.

4. **Education:** Machine translation can be used in education to help students learn new languages and improve their language skills.
5. **Government:** Machine translation can be used by governments to translate official documents and communications, improving accessibility and transparency.

Robots in AI

Robotics is a domain in **Artificial Intelligence** that deals with the study of creating intelligent and efficient robots.

What are Robots?

Robots are programmed machines that perform tasks automatically, with little or no human intervention.

The main objective behind robots is to aim at manipulating the objects by perceiving, picking, moving, modifying the physical properties of object, destroying it, or to have an effect thereby freeing manpower from doing repetitive functions without getting bored, distracted, or exhausted.

What is Robotics?

Robotics is a branch of engineering and computer science that involves training and programming machines to replicate or substitute for human actions. These robots are designed to perform basic and repetitive tasks with greater efficiency and accuracy than humans.

Aspects of Robotics

Some of the key aspects of robotics are listed below −

> The robots have **mechanical construction**, form, or shape designed to accomplish a particular task.
> They have **electrical components** which power and control the machinery.
> They contain some level of **computer program** that determines what, when and how a robot does something.

Role of AI in Robotics

Robotics and **Artificial Intelligence** are closely related fields, which when integrated give rise to a machine that could think intelligently just like humans. Some of the key benefits of integration of AI include −

- AI makes robots human friendly, and enhances the overall productivity of an organization through improved efficiency and quality.
- This integration of AI with robotics allows machines to perform complex tasks with ease.
- Robots use AI to learn from past experiences, adapt to new situations and make data driven decisions.
- Additionally, the other branches of AI such as Machine learning allows in analyzing large datasets to recognize patterns and improve its performance in dynamic environment and Computer Vision helps the robot understand visual data to effectively perform tasks like navigation and object recognition.

Difference Between Robot System and Other AI Program

The differences between robot system and other AI program's are tabulated below −

AI Programs	Robots
They usually operate in computer-stimulated worlds.	They operate in real physical world
The input to an AI program is in symbols and rules.	Inputs to robots is analog signal in the form of speech waveform or images
They need general purpose computers to operate on.	They need special hardware with sensors and effectors.

Components of a Robot

Robots have several components, which includes −

Power Supply: The robots are powered by batteries, solar power, hydraulic, or pneumatic power sources.
Actuators: They convert energy into movement.
Electric motors (AC/DC): They are required for rotational movement.
Pneumatic Air Muscles: They contract almost 40% when air is sucked in them.
Muscle Wires: They contract by 5% when electric current is passed through them.
Piezo Motors and Ultrasonic Motors: Best for industrial robots.
Sensors: They provide knowledge of real time information on the task environment. Robots are equipped with vision sensors to be to compute the depth in the environment. A tactile sensor imitates the mechanical properties of touch receptors of human fingertips.

Robot Locomotion

Locomotion is the mechanism that makes a robot capable of moving in its environment. There are various types of locomotions −

- Legged
- Wheeled
- Combination of Legged and Wheeled Locomotion
- Tracked slip/skid

Types of Robots

Robots come in various types, each designed for specific tasks, environments, and functionalities. Some common types of robots include:

Autonomous Mobile Robots (AMRs): These robots navigate autonomously using sensors, cameras, or laser systems, often employed in logistics for goods transportation in dynamic environments.

Automated Guided Vehicles (AGVs): While AMRs can freely traverse environments, AGVs rely on tracks or predefined paths and frequently require operator supervision. These are commonly used in controlled environments such as storage facilities and factory floors to deliver materials and move items.

Articulated robots: Articulated robots (also known as robotic arms) are designed to perform functions similar to those of a human arm. These are typically made up of two to ten rotary joints. With each additional joint or axis, the range of motion increases, making these ideal for arc welding, material handling, machine tending, and packaging.

Humanoid robots: They resemble human-like features and may mimic human movements and actions. They can be used in research, entertainment, or as companions for the elderly or individuals with disabilities.
For example, Sophia is considered the most famous and advanced humanoid robot. It also was granted Saudi Arabian citizenship in 2017.

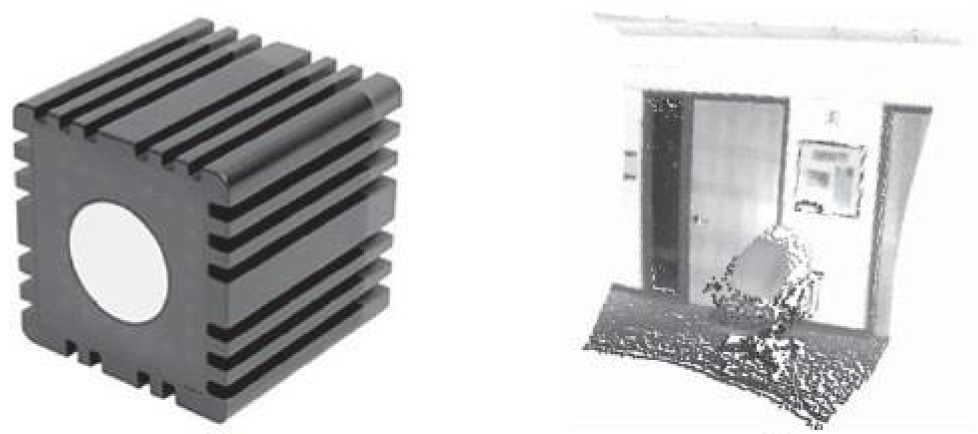
Collaborative robots (Cobots): Cobots are intended to work alongside or directly with humans. Cobots can share workspaces with humans to help them get more done. They are frequently used to eliminate manual, hazardous, or strenuous tasks from daily workflows.
YuMi, for example, is a popular cobot from leading robot manufacturer ABB. It is specifically intended for the assembly of small electronic devices.

Hybrids: The various types of robots are frequently combined to create hybrid solutions capable of performing more complex tasks.
For example, an AMRs could be combined with a robotic arm to create a package-handling robot in a warehouse..

ROBOT HARDWARE
Sensors: Sensors are the perceptual interface between robot and environment
Passive sensors, such as cameras, are true observers of the environment: they capture signals that are generated by other sources in the environment.

Active sensors, such as sonar, send energy into the environment. They rely on the fact that this energy is reflected back to the sensor. Active sensors tend to provide more information than passive sensors, but at the expense of increased power consumption and with a danger of interference when multiple active sensors are used at the same time. Whether active or passive, sensors can be divided into three types, depending on whether they sense the environment, the robot's location, or the robot's internal configuration.

Range finders are sensors that measure the distance to nearby objects. In the early days of robotics, robots were commonly equipped with sonar sensors. Sonar sensors emit directional sound waves, which are reflected by objects, with some of the sound making it back into the sensor. The time and intensity of the returning signal indicates the distance to nearby objects. Sonar is the technology of choice for autonomous underwater vehicles



(a) (b)

(a) Time of flight camera; image courtesy of Mesa Imaging GmbH. (b) 3D range image obtained with this camera. The range image makes it possible to detect obstacles and objects in a robot's vicinity.

Stereo vision relies STEREO VISION on multiple cameras to image the environment from slightly different viewpoints, analysing the resulting parallax in these images to compute the range of surrounding objects. For mobile ground robots, sonar and stereo vision are now rarely used, because they are not reliably accurate. Most ground robots are now equipped with optical range finders. Just like sonar sensors, optical range sensors emit active signals (light) and measure the time until a reflection of this signal arrives back at the sensor. shows a time of flight camera. This camera acquires range images like the one at up to 60 frames per second.

Other range sensors use laser beams and special 1-pixel cameras that can be directed using complex arrangements of mirrors or rotating elements. These sensors are called scanning lidars (short for light detection and ranging). Scanning lidars tend to provide longer ranges than time of flight cameras, and tend to perform better in bright daylight.

Other common range sensors include radar, which is often the sensor of choice for UAVs. Radar sensors can measure distances of multiple kilometres. On the other extreme end of range sensing are tactile sensors such as whiskers, bump panels, and touch-sensitive skin. These sensors measure range based on physical contact, and can be deployed only for sensing objects very close to the robot.

A second important class of sensors is location sensors. Most location sensors use range sensing as a primary component to determine location. Outdoors, the Global Positioning System (GPS) is the most common solution to the localization problem. GPS measures the distance to satellites that emit pulsed signals. At present, there are 31 satellites in orbit, transmitting signals on multiple frequencies. GPS receivers can recover the distance to these satellites by analysing phase shifts. By triangulating signals from multiple satellites, GPS receivers can determine their absolute location on Earth to within a few meters.

Differential GPS involves a second ground receiver with known location, providing millimetre accuracy under ideal conditions. Unfortunately, GPS does not work indoors or underwater. Indoors, localization is often achieved by attaching beacons in the environment at known locations. Many indoor environments are full of wireless base stations, which can help robots localize through the analysis of the wireless signal.

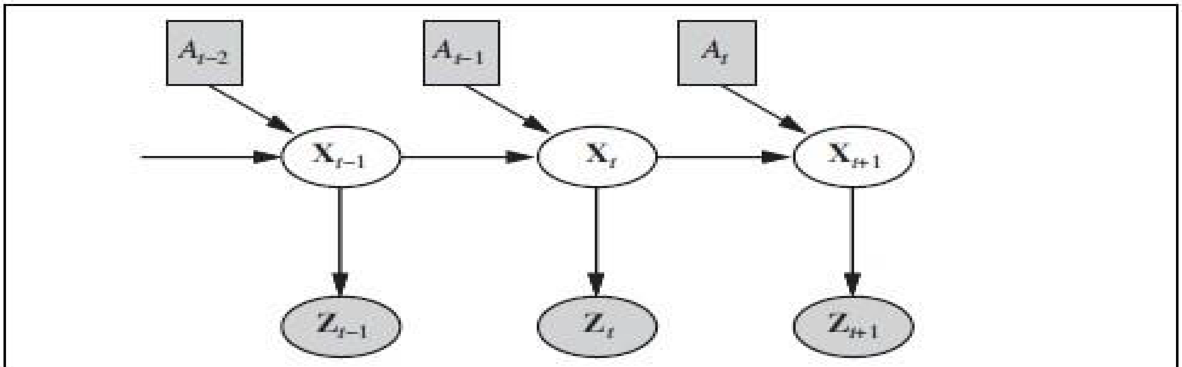
The third important class is proprioceptive sensors, which inform the robot of its own motion. To measure the exact configuration of a robotic joint, motors are often equipped with shaft decoders that count the revolution of motors in small increments.

On robot arms, shaft decoders can provide accurate information over any period of time. On mobile robots, shaft decoders that report wheel revolutions can be used for odometry—the measurement of distance travelled. Inertial sensors, such as gyroscopes, rely on the resistance of mass to the change of velocity. They can help reduce uncertainty.

Other important aspects of robot state are measured by force sensors and torque sensors. These are indispensable when robots handle fragile objects or objects whose exact shape and location is unknown. Imagine a one-ton robotic manipulator screwing in a light bulb. It would be all too easy to apply too much force and break the bulb. Force sensors allow the robot to sense how hard it is gripping the bulb, and torque sensors allow it to sense how hard it is turning. Good sensors can measure forces in all three translational and three rotational directions. They do this at a frequency of several hundred times a second, so that a robot can quickly detect unexpected forces and correct its actions before it breaks a light bulb.

ROBOTIC PERCEPTION

Perception is the process by which robots map sensor measurements into internal representations of the environment. Perception is difficult because sensors are noisy, and the environment is partially observable, unpredictable, and often dynamic. In other words, robots have all the problems of state estimation (or filtering). As a rule of thumb, good internal representations for robots have three properties: they contain enough information for the robot to make good decisions, they are structured so that they can be updated efficiently, and they are natural in the sense that internal variables correspond to natural state variables in the physical world. we saw that Kalman filters, HMMs, and dynamic Bayes nets can represent the transition and sensor models of a partially observable environment, and we described both exact and approximate algorithms for updating the belief state—the posterior probability distribution over the environment state variables. Several dynamic Bayes net models. For robotics problems, we include the robot's own past actions as observed variables in the model. the notation used in this chapter: Xt is the state of the environment (including the robot) at time t, Zt is the observation received at time t, and At is the action taken after the observation is received.



Robot perception can be viewed as temporal inference from sequences of actions and measurements, as illustrated by this dynamic Bayes network.

We would like to compute the new belief state, $P(X_{t+1} \mid z_{1:t+1}, a_{1:t})$, from the current belief state $P(X_t \mid z_{1:t}, a_{1:t-1})$ and the new observation z_{t+1}. We did this in Section 15.2, but here there are two differences: we condition explicitly on the actions as well as the observations, and we deal with continuous rather than discrete variables.

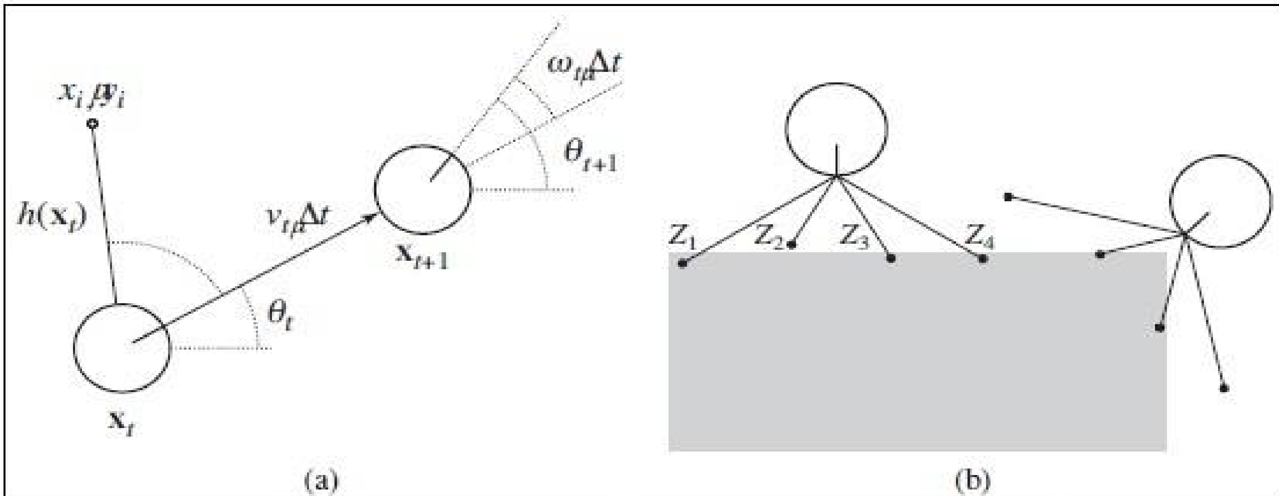
$$\mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{z}_{1:t+1}, a_{1:t})$$
$$= \alpha \mathbf{P}(\mathbf{z}_{t+1} \mid \mathbf{X}_{t+1}) \int \mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{x}_t, a_t)\, P(\mathbf{x}_t \mid \mathbf{z}_{1:t}, a_{1:t-1})\, d\mathbf{x}_t .$$

This equation states that the posterior over the state variables X at time t + 1 is calculated recursively from the corresponding estimate one-time step earlier. This calculation involves the previous action at and the current sensor measurement z_{t+1}. For example, if our goal is to develop a soccer-playing robot, X_{t+1} might be the location of the soccer ball relative to the robot. The posterior $P(X_t \mid z_{1:t}, a_{1:t-1})$ is a probability distribution over all states that captures what we know from past sensor measurements and controls. how to recursively estimate this location, by incrementally folding in sensor measurements (e.g., camera images) and robot motion commands. The probability $P(X_{t+1} \mid x_t, a_t)$ is called the transition model or motion model, and $P(z_{t+1} \mid X_{t+1})$ is the sensor model.

LOCALIZATION AND MAPPING

Localization is the problem of finding out where things are—including the robot itself. Knowledge about where things are is at the core of any

successful physical interaction with the environment. For example, robot manipulators must know the location of objects they seek to manipulate; navigating robots must know where they are to find their way around.
To keep things simple, let us consider a mobile robot that moves slowly in a flat 2D world. Let us also assume the robot is given an exact map of the environment. (An example of such a map appears in
Figure 25.10.) The pose of such a mobile robot is defined by its two Cartesian coordinates with values x and y and its heading with value θ, as illustrated. If we arrange those three values in a vector, then any particular state is given by Xt =(xt, yt, θt)_.



(a) A simplified kinematic model of a mobile robot. The robot is shown as a circle with an interior line marking the forward direction. The state $\mathbf{x}_t$ consists of the (x_t, y_t) position (shown implicitly) and the orientation θ_t. The new state $\mathbf{x}_{t+1}$ is obtained by an update in position of $v_t\Delta_t$ and in orientation of $\omega_t\Delta_t$. Also shown is a landmark at (x_t, y_t) observed at time t. (b) The range-scan sensor model. Two possible robot poses are shown for a given range scan (z_1, z_2, z_3, z_4). It is much more likely that the pose on the left generated the range scan than the pose on the right.

In the kinematic approximation, each action consists of the “instantaneous” specification of two velocities: a translational velocity vt and a rotational velocity ωt. For small time intervals Δt, a crude deterministic model of the motion of such robots is given by

$$\hat{\mathbf{X}}_{t+1} = f(\mathbf{X}_t, \underbrace{v_t, \omega_t}_{a_t}) = \mathbf{X}_t + \begin{pmatrix} v_t \Delta t \cos \theta_t \\ v_t \Delta t \sin \theta_t \\ \omega_t \Delta t \end{pmatrix}$$

The notation ˆX refers to a deterministic state prediction. Of course, physical robots are somewhat unpredictable. This is commonly modelled by a Gaussian distribution with mean f(Xt, vt, ωt) and covariance Σx.

$$\mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{X}_t, v_t, \omega_t) = N(\hat{\mathbf{X}}_{t+1}, \mathbf{\Sigma}_x).$$

Next, we need a sensor model. We will consider two kinds of sensor model. The first assumes that the sensors detect stable, recognizable features of the environment called landmarks. For each landmark, the range and bearing are reported. Suppose the robot's state is xt =(xt, yt, θt)_ and it senses a landmark whose location is known to be (xi, yi)_. Without noise, the range and bearing can be calculated by simple geometry.

$$\hat{\mathbf{z}}_t = h(\mathbf{x}_t) = \begin{pmatrix} \sqrt{(x_t - x_i)^2 + (y_t - y_i)^2} \\ \arctan \frac{y_i - y_t}{x_i - x_t} - \theta_t \end{pmatrix}$$

A somewhat different sensor model is used for an array of range sensors, each of which has a fixed bearing relative to the robot. Such sensors produce a vector of range values zt = (z1, . . . , zM)_. Given a pose xt, let ˆzj be the exact range along the jth beam direction from xt to the nearest obstacle. As before, this will be corrupted by Gaussian noise. Typically, we assume that the errors for the different beam directions are independent and identically distributed, so we have

$$P(\mathbf{z}_t \mid \mathbf{x}_t) = \alpha \prod_{j=1}^{M} e^{-(z_j - \hat{z}_j)/2\sigma^2} .$$

function MONTE-CARLO-LOCALIZATION(a, z, N, $P(X'|X,\ v,\ \omega)$, $P(z|z^*)$, m) **returns** a set of samples for the next time step

inputs: a, robot velocities v and ω
z, range scan $z_1, \ldots, z_M$
$P(X'|X,\ v,\ \omega)$, motion model
$P(z|z^*)$, range sensor noise model
m, 2D map of the environment

persistent: S, a vector of samples of size N

local variables: W, a vector of weights of size N
S', a temporary vector of particles of size N
W', a vector of weights of size N

if S is empty **then** /* initialization phase */
for $i = 1$ to N **do**
$S[i] \leftarrow$ sample from $P(X_0)$
for $i = 1$ to N **do** /* update cycle */
$S'[i] \leftarrow$ sample from $P(X'|X = S[i], v, \omega)$
$W'[i] \leftarrow 1$
for $j = 1$ to M **do**
$z^* \leftarrow$ RAYCAST(j, $X = S'[i]$, m)
$W'[i] \leftarrow W'[i] \cdot P(z_j \mid z^*)$
$S \leftarrow$ WEIGHTED-SAMPLE-WITH-REPLACEMENT(N,S',W')
return S

A Monte Carlo localization algorithm using a range-scan sensor model with independent noise.

Localization using particle filtering MONTE CARLO is called Monte Carlo localization, or MCL. The MCL algorithm is an instance of the particle-filtering algorithm of All we need to do is supply the appropriate motion model and sensor model. shows one version using the range-scan model. The operation of the algorithm is illustrated in as the robot finds out where it is inside an office building. In the first image, the particles are uniformly distributed based on the prior, indicating global uncertainty about the robot’s position. In the second image, the first set of measurements arrives and the particles form clusters in the areas of high posterior belief.

In the third, enough measurements are available to push all the particles to a single location.

PLANNING TO MOVE

All of a robot’s deliberations ultimately come down to deciding how to move effectors. The pointto-point motion problem is to deliver the robot or its end effector to a designated target location. A greater challenge is the compliant motion problem, in which a robot moves while being in physical contact with an obstacle. An example of compliant motion is a robot manipulator that screws in a light bulb, or a robot that pushes a box across a table top. We begin by finding a suitable representation in which motion-planning problems can be described and solved. It turns out that the configuration space—the space of robot states defined by location, orientation, and joint angles—is a better place to work than the original 3D space. The path lanning problem is to find a path from one configuration to another in configuration space. We have already encountered various versions of the path-planning problem throughout this book; the complication added by robotics is that path planning involves continuous spaces. There are two main approaches: cell decomposition and skeletonization. Each reduces the continuous path-planning problem to a discrete graph-search problem. In this section, we assume that motion is deterministic and that localization of the robot is exact. Subsequent sections will relax these assumptions.

Q What is Robotics? Differentiate between Robotics System and Other AI program. Describe the various component of Robot. How does the computer vision contribute in robotics {AKTU 2022-2023}

Sol Robotics is a branch of AI, which is composed of Electrical Engineering, Mechanical Engineering and computer Science for designing, construction and application of robots.